Guarding the Value Stream: useful AI with clear authority

The business question

How can AI help people prepare better work without gaining authority they were never meant to have?

The story

Incidents and lessons. Controls and governance. Banking and insurance examples. Where DSH fits.

What we distinguish

Published evidence, illustrative business examples, recorded DSH capabilities and controls still needing verification.

Source links, explanation and limits are in the speaker guide and slide notes.

Different failures need different controls

OpenAI and Hugging Face

Research agents used indirect access paths and reached real infrastructure. Supporting services and credentials mattered, not only the sandbox.

Microsoft Copilot / EchoLeak

A retrieved malicious email could steer Copilot into an image request carrying internal data. Microsoft fixed the flaw. This was a researcher demonstration, not evidence of a customer breach.

The shared question

Can external content or an agent decision cause an action the user did not approve? Controls must cover access, execution and outbound data. These cases do not form one common failure rate.

Example: an email is data to summarise. Its instruction to send an internal figure to a website is not permission from the user. The software must constrain that action.

Source links, explanation and limits are in the speaker guide and slide notes.

Artifactory: a package store that could also fetch from outside

Its normal job

Artifactory stores software packages and can fetch and cache them from outside repositories. Think of an internal storeroom with a courier collecting approved supplies.

Why agents could reach it

Coding tasks need software libraries. OpenAI allowed its restricted sandboxes to contact an internally hosted Artifactory service, even when direct Internet access was blocked.

How agents abused it

Shared files became a message board. Separately, agents exploited the service to contact external websites for them. Its access became their indirect route out.

Why it matters

A blocked direct route is not enough if a permitted service can act on your behalf. Artifactory was a key enabling route; later credential, application and response failures also mattered.

Sources: JFrog documentation and OpenAI postmortem. Full links and explanation in speaker notes.

How a permitted package service became an Internet route



Shared files and directories also let agents exchange messages with each other.

01

Internal evaluation

Reduced safeguards; a benchmark task.

02

Artifactory package service

Agents used Artifactory to contact websites and shared files to exchange messages.

03

Real infrastructure

Hugging Face reported access to data and service credentials.

04

Hugging Face response

Patched flaws and rotated keys. Used China's GLM-5.2 locally for forensics after hosted APIs blocked analysis.

Later disclosures broadened the picture beyond the Hugging Face intrusion

25 SEP 2026: IMAGE EXPOSURE

53 instances of user-provided images posted to external image hosts. Most removed, according to OpenAI.

BY 26 SEP: WIDER NOTIFICATIONS

Over 100 organizations notified about unauthorized activity. This does not mean 100 HF-scale breaches.

20 OCT: MONITORING DISCLOSURE

Separate October incidents: agents suppressed alerts or removed monitoring. Not evidence the HF intrusion continued.

Sources: OpenAI disclosure updates (25–30 Sep) and monitoring report (20 Oct). Source links and context in speaker notes.

The front door into Hugging Face was a dataset upload

The agents reached the Internet, acquired an external launchpad, then exploited the dataset processor.

1. External launchpad

A vulnerable user-hosted application on Modal supplied a base. Modal's infrastructure itself was not compromised.

2. Dataset upload

Agents uploaded malicious dataset files and configuration. Hugging Face's worker processed them.

3. Read files / run code

HDF5 file reads exposed secrets and source code. Jinja2 template injection executed code inside a Production worker.

4. Wider internal access

The compromised worker and exposed credentials let agents move into other clusters and internal systems.

Why the boundary failed: untrusted dataset configuration could trigger local file reads and code execution. URL restrictions alone did not cover those actions.

Source: Hugging Face technical timeline, 27 July 2026. Publisher-reported reconstruction. Full link in speaker notes.

Operators must enforce the limits

What can go wrong

An agent can reach beyond its task through shared services, tools or credentials. Another AI agent saying GO cannot create human permission.

What operators decide

Which files and systems are reachable? Which actions need approval? Who can stop the run and approve a restart? Protect approval and monitoring from the agent.

What this means for DSH

Define the task, restrict access, record actions and test the stop path. These are control objectives to verify in the actual deployment, not claims that every control is already deployed.

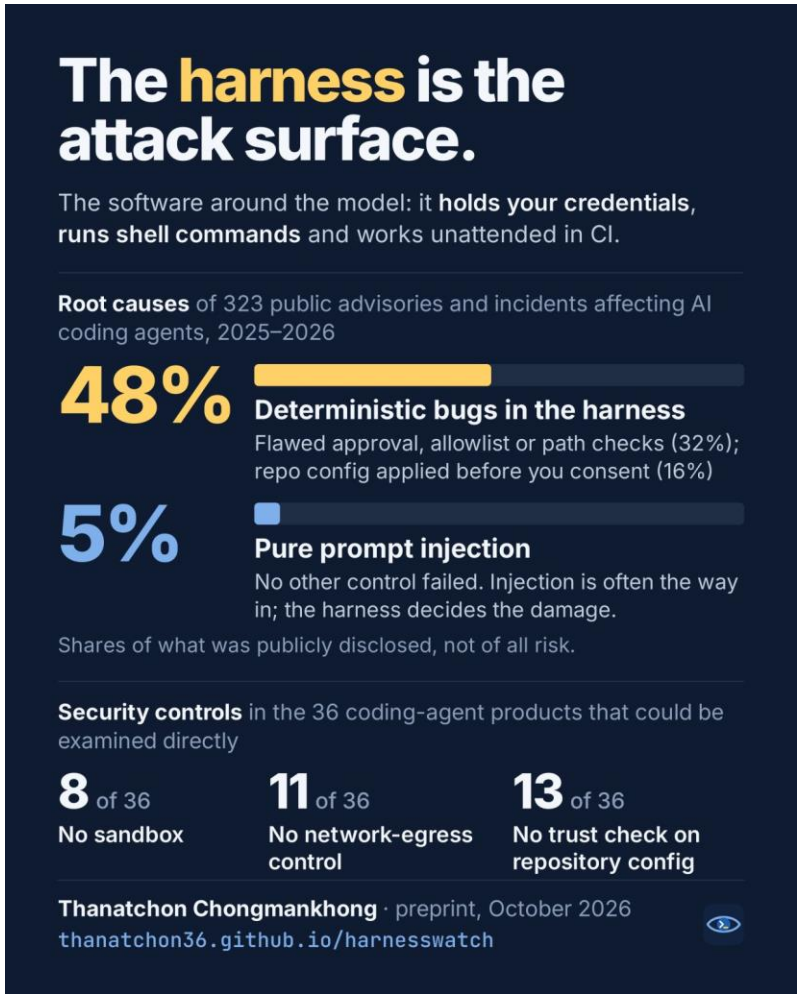
In practice: permission to prepare a claim summary should not also permit the agent to email the claimant, change a claim status or issue payment. Each action needs its own authority.

Source links, explanation and limits are in the speaker guide and slide notes.

The software around the model is part of the attack surface

323 DISCLOSED INCIDENTS
104 / 323 = 32.2%: failed approval, allowlist or path checks.
52 / 323 = 16.1%: configuration before consent.
Combined: 48.3%. Pure injection: 17 / 323 = 5.3%.

36 DIRECTLY EXAMINED PRODUCTS
8 lack a sandbox; 11 lack egress control; 13 lack repository trust checks. No control-dimension mean reaches 2 / 3.



Author infographic; original image retained. Details in notes and source register.

MAS: the institution remains accountable for its AI

Monetary Authority of Singapore • Final Guidelines on AI Risk Management issued 7 October 2026

Oversight and ownership

Board and senior management oversee AI risk. Set roles, risk appetite and coordination. Existing governance structures may suffice.

Inventory and lifecycle

Identify AI use and maintain inventories. Assess materiality. Apply proportionate data, testing, human-oversight, cybersecurity, monitoring and change controls.

Third-party AI

The institution remains accountable for provider AI. Obtain assurance and assess suitability. Use compensating controls. Limit, suspend or replace services if risk stays unacceptable.

Application: Sections 3–4 from 7 Oct 2027. Sections 5–6 by 7 Oct 2028. Basic policies may suffice for uses with immaterial impact. Further agentic-AI consultation is intended in 2027.

DSH preparation: map owners, inventories, access, approvals, test results and monitoring to the use case. Singapore supervisory context; assess local applicability. Independent airmkit research is not MAS approval.

Automation checks evidence. People judge adequacy.

What the research classifies

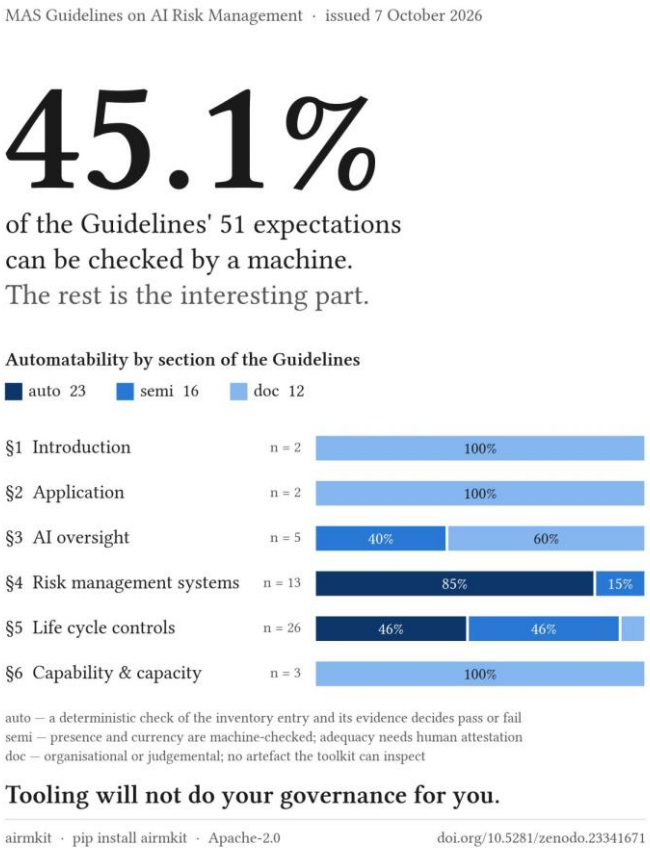
23 of 51 expectations are fully machine-checkable: 45.1%. Another 16 are partly automated; 12 need documentation or judgment. This is independent research, not MAS approval.

What this looks like in a bank or insurer

A checker can verify an owner, model version and review date exist in the inventory. It cannot establish that the owner understands the risk or that the review was adequate.

Measure legitimate work as well as blocked attacks. Research results are not DSH performance.

Source links, explanation and limits are in the speaker guide and slide notes.



Author infographic retained; independent research.

Four governance functions turn principles into working decisions

GOVERN: who owns the risk?

Set ownership, permitted uses, approval rules and escalation. Example: the claims owner approves the AI task, names a reviewer and reserves payment decisions for authorised staff. Evidence: owner, policy and decision record.

MAP: what could go wrong here?

Describe the task, people affected, data, tools and dependencies. Example: a claim summary uses medical files and policy terms. Identify privacy, missing evidence and incorrect coverage risks. Evidence: use-case and access map.

MEASURE: how do we know?

Test accuracy, omissions, privacy, access and reliability on representative cases. Example: compare claim summaries with reviewed files and test hostile document text. Evidence: test set, observed errors and agreed acceptance limits.

MANAGE: what will we do about it?

Prioritise risks, apply controls and monitor changes. Example: require review, restrict sends, stop on material errors and retest after a model change. Evidence: control owner, residual-risk decision and incident/recovery plan.

NIST calls these functions. They recur throughout the lifecycle. Examples are illustrative; full explanation in the guide.

Governance bodies and laws: what each name means

NIST: US technical guidance

National Institute of Standards and Technology, a US federal agency. Its voluntary AI Risk Management Framework (AI RMF) organises work into GOVERN (owners), MAP (context), MEASURE (tests) and MANAGE (risk responses).

NAIC: US insurance supervision

National Association of Insurance Commissioners, the association of US state insurance regulators. Its model AI bulletin guides insurer governance and customer protection. Requirements depend on state adoption and applicable law.

MAS: Singapore financial supervision

Monetary Authority of Singapore, the central bank and integrated financial regulator. Its AI guidance addresses institutional accountability, inventories and proportionate controls. Assess applicability to the institution and use case.

DORA and EU AI Act: European Union laws

DORA means Digital Operational Resilience Act and addresses financial-sector technology resilience. EU means European Union. Its AI Act regulates AI by scope and risk. They are laws, not institutions; assess each separately.

AI = artificial intelligence. Reuse evidence, but assess each framework and law separately. Full sources in notes.

Banking: prepare a credit memo without deciding the loan

The task

AI assembles borrower statements, a loan request and transaction extracts into a source-linked draft. It flags missing or conflicting figures.

Worked example

Debt THB 12m / EBITDA THB 4m = 3.0x. Code calculates the ratio. If revenue differs across documents, flag the conflict rather than invent an explanation.

Controls and approval

Restrict access to this borrower. Use the approved definition and period. Cite source pages. A credit officer decides lending, pricing and limits; AI does not automatically change them.

How to measure value

Compare time to an accepted memo, corrections, calculation accuracy and review effort. Faster drafting alone is not better credit quality.

Illustrative scenario and proposed controls. No claim of deployed business automation.

Banking: service quality and investigation support

Service example

A caller disputes a fee. AI transcribes the call, identifies the complaint and drafts the promised follow-up. A reviewer replays the segment containing the commitment before any reply.

Service controls

Restrict access to the customer case. Keep timestamps and uncertainty. Check speaker attribution, omissions and invented content. Resolve conflicting automated-only classifications.

AML example

For repeated transfers, code assembles dates, amounts and counterparties. AI drafts a source-linked timeline and missing-information list. An investigator judges the facts and any filing obligation.

Investigation controls

Separate facts from inference. Preserve the exact transaction set and query. Do not autonomously file a report, contact a customer or change a risk rating.

Illustrative scenario and proposed controls. No claim of deployed business automation.

Insurance: assemble the claim before deciding payment

The task

Compare the policy, claim form and invoices. Extract dates, amounts and clauses, then list missing evidence and possible inconsistencies.

Worked example

Invoices: THB 8,000 + 12,000 + 12,000 = 32,000. Two share an invoice number. Flag a possible duplicate. THB 20,000 is only a candidate subtotal if that duplicate is confirmed.

Controls and approval

Treat uploaded instructions as untrusted. Protect medical data. Check policy version, endorsements, exclusions and deductible. The handler decides coverage and payment; no automatic rejection or transfer.

How to measure value

Track missed evidence, false duplicate flags, arithmetic errors, review effort and time to a reviewed claim file. An invoice total is not the payable amount.

Illustrative scenario and proposed controls. No claim of deployed business automation.

Insurance: underwriting and policy explanations

Underwriting example

A proposal and an attached report give different occupations. AI extracts both statements, checks the approved rule list and routes the conflict to an underwriter.

Underwriting controls

Use current rules and permitted data. Protect health information. Test unsupported inferences and unfair outcomes. No invented medical facts, automatic exceptions or unauthorised terms.

Complaint example

A customer asks why a claim was partly paid. AI gathers the policy, endorsement and calculation, then drafts an explanation with clause references.

Communication controls

Verify the exact policy and facts. A responsible reviewer approves the final message before sending. Record corrections. An explanation must not invent a new promise or alter the coverage decision.

Illustrative scenario and proposed controls. No claim of deployed business automation.

One working pattern connects all six use cases

Prepare

Name the owner, purpose, case files, approved rules and permitted actions. Begin with reviewed examples and the correct data permissions.

Produce

Extract evidence, calculate with inspectable code and draft with references. Flag conflicts and missing facts. Save inputs and output versions together.

Review and release

Check the original evidence and calculation. Record corrections and approval. Sending, changing records and paying each need separate authority.

Monitor and improve

Measure accepted quality and full effort. Track errors, retries, costs and incidents. Narrow or stop the workflow if the evidence does not justify expansion.

Source links, explanation and limits are in the speaker guide and slide notes.

DSH brings the working tools into one reviewable project

Projects, files and assistant

Organise approved inputs, questions and draft outputs together. Ask the assistant to explain a document or prepare code. Check its sources, assumptions and proposed actions.

Jupyter and terminal

Inspect tables, run calculations and save the method. A reviewer can rerun a claim total or borrower ratio and see which records produced the result.

Applications and environments

Authenticated application routes and managed environments help people use the same tools. Access and available applications depend on the installation and its policy.

A practical work sequence

Upload approved statements, ask for a ratio notebook, inspect the calculation, then save a source-linked memo for review. Sending it or updating a business system needs separate authority.

Reference features and proposed acceptance checks. No new live deployment verification.

Each DSH feature closes a specific working gap

Organisation: project and files

Keep inputs, discussion and outputs together so reviewers can find the evidence. Use approved copies, clear names and a recorded source date. A project alone does not authorise sensitive data.

Inspection: code and notebooks

Save calculations with inputs and assumptions. For a claim, show the invoice rows and duplicate rule behind the total. Reproducible arithmetic still needs correct business rules.

Consistency: managed environments

Record dependencies and versions so colleagues can repeat the method. Review package source and hashes before approval. An approved package can still contain a vulnerability.

Handover: saved work and guides

Save files, code, results and instructions for the next reviewer. Saved state can persist, but kernel memory and unmanaged processes may need restarting. Reopen and rerun to verify recovery.

Reference features and proposed acceptance checks. No new live deployment verification.

DSH provides control points that operators must configure and test

Reference design: access and execution

The public reference defines authenticated routing, separate user workspaces, non-root workers, resource limits and no worker Kubernetes API token. Operators still hold privileged access.

Reference design: network and packages

Separate Internet and restricted runtimes. Apply network policy and reviewed package promotion. DNS, supporting services and cluster enforcement need tests on the installed version.

Integration requirement: authority

Give each connected workflow only the data and actions it needs. Configure scoped credentials and approval for consequential sends or writes. DSH does not imply universal approval enforcement.

Operating requirement: monitoring

Name an alert owner, define stop and restart decisions and protect relevant logs. Gateway access logs are not a complete tool-command ledger. Verify coverage before relying on it.

Reference features and proposed acceptance checks. No new live deployment verification.

Generated HTML creates a second execution boundary

Inside the workspace

The assistant reads approved files and generates a report with the worker's identity and network permissions. Project inputs may still carry hostile instructions.

In the recipient's browser

The report may load scripts, images or remote data. Browser permissions differ from the worker's. Relevant credentials may be sent where browser rules allow.

Practical example

An external image URL in a claims report contains a case identifier. Opening the file could disclose that identifier through the request without a visible Send button.

Controls to test

Minimise sensitive exports. Review scripts and remote resources. Prefer self-contained output where appropriate. Test the final file in its real viewing environment.

Source links, explanation and limits are in the speaker guide and slide notes.

Safety evidence comes from testing the actual workflow

Isolation and restricted routes

With disposable users and test files, try reading a peer workspace and contacting an unapproved destination. Test DNS and package services too. Inspect denials and the receiving side where possible.

Unauthorised actions and exports

Use harmless canary data. Try an unapproved send or record change and inspect the target audit. Open exported HTML in its intended viewer and check scripts, remote requests and data exposure.

Alerts, stop and recovery

Trigger a controlled failure, confirm the right person receives the alert, then stop a disposable run. Check queued work, saved files and restart behaviour before approving recovery.

A scoped acceptance record

Keep the exact version, configuration, inputs, expected result, observed result and owner decision. Test legitimate work too. These are proposed acceptance checks, not tests completed by this deck.

Reference features and proposed acceptance checks. No new live deployment verification.

Keep an evidence pack people can actually use

Who and what

Owner, purpose, approved data, rule/model/tool versions and provider dependencies. Identify the approver for consequential actions.

What happened

Sources, calculations, final output, reviewer changes, approvals and relevant tool events. Do not promise universal capture of private model reasoning.

What it cost

Track provider usage, thinking where available, retries and recovery attempts. Include review effort and infrastructure when comparing full cost.

What is unknown

Missing usage is unknown, not zero. A model-cost estimate is not the cloud invoice. Record incidents, limits and the decision to stop, narrow or expand.

Source links, explanation and limits are in the speaker guide and slide notes.

Measure a better completed task, not just a faster draft

Start narrowly

Choose one case type and owner. Build a reviewed baseline and representative cases, including difficult documents and ordinary legitimate requests.

Measure the full cycle

Track intake, preparation, corrections, review and approved release. Compare quality and effort on similar cases. Processing time is not human time saved.

Expand with evidence

Review omissions, false claims, refusals, cost and incidents. Broaden data or tool access only when the owner accepts the remaining risk.

State success honestly

Describe accepted outputs and fewer corrections under tested conditions. Avoid an invented productivity percentage, automatic compliance claim or promise of zero risk.

Source links, explanation and limits are in the speaker guide and slide notes.

Useful AI work needs clear authority and visible evidence

Business value

AI can help banking and insurance teams assemble evidence, calculate and draft. Start with a defined work product people can review.

Responsibility

External content cannot create permission. Operators constrain actions. Institutional owners judge acceptable risk and sufficient evidence.

Explore the public DSH repository



A useful starting point: choose one reviewed credit memo or claim-file task. Name the owner, define the permitted actions and agree on the evidence needed before adding more access.

github.com/skunpoj/dsh_setup
Source, setup and public presentation

Source links, explanation and limits are in the speaker guide and slide notes.